

CHAPTER 3

BIOENERGETICS OF EXERCISE AND TRAINING

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After completing this chapter, you will be able to

- explain the basic energy systems available to supply ATP during exercise;
- understand lactate accumulation, metabolic acidosis, and cellular manifestations of fatigue;
- identify patterns of substrate depletion and repletion during various exercise intensities;
- describe the bioenergetic factors that limit exercise performance;
- critique training programs based on the metabolic specificity of training; and
- explain the metabolic demands of and recovery from interval training, high-intensity interval training, and combination training to optimize work-to-rest ratios.

Metabolic specificity of exercise and training is based on an understanding of the transfer of energy in biological systems.

Through an understanding of how energy transfer occurs, more efficient and productive training programs can be designed. After defining essential bioenergetics terminology and explaining the role of adenosine triphosphate (ATP), this chapter discusses the three basic energy systems that work to replenish ATP in human skeletal muscle. Then we look at substrate depletion and repletion, especially as they relate to fatigue and recovery; bioenergetic factors that limit performance; and aerobic and anaerobic contributions to oxygen uptake. Finally, the metabolic specificity of training is discussed.

Essential Terminology

Bioenergetics, or the flow of **energy** in a biological system, concerns primarily the conversion of macronutrients—carbohydrate, protein, and fats, which contain chemical energy—into biologically usable forms of energy. It is the breakdown of the chemical bonds in these macronutrients that provides the energy necessary to perform biological work.

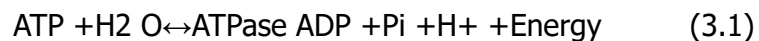
The breakdown of large molecules into smaller molecules, associated with the release of energy, is termed **catabolism**. The synthesis of larger molecules from smaller molecules can be accomplished using the energy released from catabolic reactions; this building-up process is termed **anabolism**. The breakdown of protein into amino acids is an example of catabolism, while the formation of proteins from amino acids is an anabolic process.

Exergonic reactions are energy-releasing reactions and are generally catabolic. **Endergonic reactions** require energy and include anabolic processes and the contraction of muscle.

Metabolism is the sum of all catabolic or exergonic and anabolic or endergonic reactions in a biological system. Energy derived from catabolic or exergonic reactions is used to drive anabolic or endergonic reactions through an intermediate molecule, **adenosine triphosphate (ATP)**. ATP allows the transfer of energy from exergonic to endergonic reactions. Thus, exergonic (i.e., energy-releasing) reactions are coupled with endergonic (i.e., energy-requiring) reactions, based on the production or breakdown of ATP, respectively. Without an adequate supply of ATP, muscular activity and muscle growth would not be possible because the process reflects the metabolic system's current ability to supply the demand of the required activity. Thus, when an exercise steady state is achieved, ATP supply meets ATP demand. However, fatigue results when ATP supply does not meet the ATP requirements demanded by a given activity. Following a period of appropriate conditioning, the metabolic systems of the body adapt to be better able to supply the

ATP to meet the ATP demand of the same activity that once resulted in fatigue. It is important to note that the nature of the activity has not changed, but the metabolic system's capacity to do the activity changes with training. Thus, when designing training programs, strength and conditioning professionals need to have a basic understanding of how exercise affects ATP hydrolysis and resynthesis.

Adenosine triphosphate is composed of adenosine and three phosphate groups ([figure 3.1](#)). Adenosine is the combination of adenine (a nitrogen-containing base) and ribose (a five-carbon sugar). The breakdown of one molecule of ATP to yield energy is known as **hydrolysis**, because it requires one molecule of water. The hydrolysis of ATP is catalyzed by the presence of an enzyme called **adenosine triphosphatase (ATPase)**. Specifically, **myosin ATPase** is the enzyme that catalyzes ATP hydrolysis for crossbridge recycling. Other specific enzymes hydrolyze ATP at other locations, such as **calcium ATPase** for pumping calcium into the sarcoplasmic reticulum (following muscle contraction) and **sodium-potassium ATPase** for maintaining the sarcolemmal concentration gradient after depolarization (59). The following equation depicts the reactants (left), enzyme (middle), and products (right) of ATP hydrolysis:



where ADP represents **adenosine diphosphate** (only two phosphate groups, [figure 3.1](#)), P_i is an **inorganic phosphate** molecule, and H^+ is a hydrogen ion (proton). Further hydrolysis of ADP cleaves the second phosphate group and yields **adenosine monophosphate (AMP)**. The energy released primarily from the hydrolysis of ATP and secondarily from ADP results in biological work.

ATP is classified as a high-energy molecule because it stores large amounts of energy in the chemical bonds of the two terminal phosphate groups. Because muscle cells store ATP only in limited amounts and activity requires a constant supply of ATP to provide

the energy needed for muscle actions, ATP-producing processes must occur in the cell.

Biological Energy Systems

Three basic energy systems exist in mammalian muscle cells to replenish ATP (87, 124):

- Phosphagen
- Glycolytic
- Oxidative

In discussion of exercise-related bioenergetics, the terms *anaerobic* and *aerobic metabolism* are often used. **Anaerobic** processes do not require the presence of oxygen, whereas **aerobic** mechanisms depend on oxygen. The **phosphagen** and **glycolytic systems** are anaerobic mechanisms that occur in the sarcoplasm of a muscle cell. The **Krebs cycle**, electron transport, and the rest of the **oxidative system** are aerobic mechanisms that occur in the **mitochondria** of muscle cells and require oxygen as the terminal electron acceptor.

Of the three main macronutrients—carbohydrate, protein, and fats—only carbohydrate can be metabolized for energy without the direct involvement of oxygen (21). Therefore, carbohydrate is critical during anaerobic metabolism. All three energy systems are active at any given time; however, the magnitude of the contribution of each system to overall work performance is primarily dependent on the intensity of the activity and secondarily on the duration (45, 87).

Energy stored in the chemical bonds of adenosine triphosphate (ATP) is used to power muscular activity. The replenishment of ATP in human skeletal muscle is accomplished by three basic energy systems: (a) phosphagen, (b) glycolytic, and (c) oxidative. The first few seconds of activity are supplied by ATP stores before these metabolic energy systems begin the process of replenishing ATP stores and meeting the activity demands.

Phosphagen System

The phosphagen system provides ATP primarily for short-term, high-intensity activities (e.g., resistance training and sprinting) and is highly active at the start of all exercise regardless of intensity (62, 70, 155). This energy system relies on the hydrolysis of ATP (equation 3.1) and breakdown of another high-energy phosphate molecule called **creatine phosphate (CP)**, also called **phosphocreatine (PCr)**. **Creatine kinase** is the enzyme that catalyzes the synthesis of ATP from CP and ADP in the following reaction:



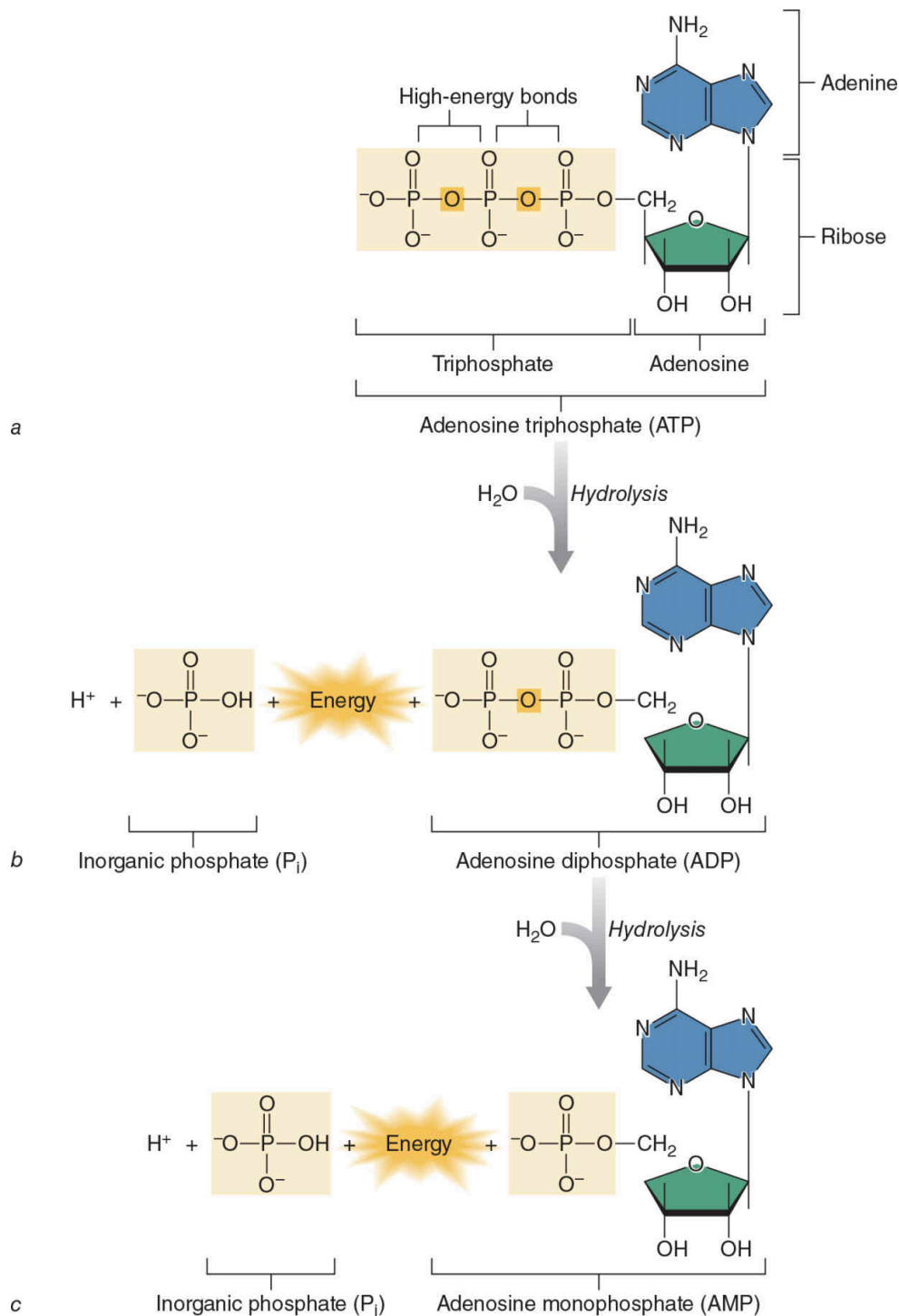


FIGURE 3.1 (a) The chemical structure of an ATP molecule showing adenosine (adenine + ribose), the triphosphate group, and locations of the high-energy chemical bonds. (b) The hydrolysis of ATP breaks the terminal phosphate bond, releases energy, and leaves ADP, a P_i , and a hydrogen ion (H^+). (c) The hydrolysis of ADP breaks the terminal phosphate bond, releases energy, and leaves AMP, P_i , and H^+ .

CP supplies a phosphate group that combines with ADP to replenish ATP. The creatine kinase reaction provides energy at a high rate; however, because CP is stored in relatively small amounts, the phosphagen system cannot be the primary supplier of energy for continuous, long-duration activities (30).

ATP Stores

The body stores approximately 80 to 100 grams (about 3 oz) of ATP at any given time, which does not represent a significant energy reserve for exercise (109). In addition, ATP stores cannot be completely depleted due to the necessity for basic cellular function. In fact, ATP concentrations may only decrease by up to 50% to 60% (34, 71, 102, 145) of the pre-exercise levels during experimentally induced muscle fatigue. Therefore, the phosphagen system uses the creatine kinase reaction (equation 3.2) to maintain the ATP concentration. Under normal circumstances, skeletal muscle concentrations of CP are four to six times higher than ATP concentrations (109). Therefore, the phosphagen system, through CP and the creatine kinase reaction, serves as an energy reserve for rapidly replenishing ATP. In addition, type II (fast-twitch) muscle fibers contain higher concentrations of CP than type I (slow-twitch) fibers (95, 134); thus, individuals with higher percentages of type II fibers may be able to replenish ATP more quickly through the phosphagen system during anaerobic, explosive exercise.

Another important single-enzyme reaction that can rapidly replenish ATP is the **adenylate kinase** (also called **myokinase**) **reaction**:



This reaction is particularly important because AMP, a product of the adenylate kinase (myokinase) reaction, is a powerful stimulant of **glycolysis** (22, 28).

Control of the Phosphagen System

The phosphagen system reactions (often represented by equations 3.1, 3.2, and 3.3) are largely controlled by the **law of mass action** or the **mass action effect** (109). The law of mass action states that the concentrations of reactants or products (or both) in solution will drive the direction of the reactions. With enzyme-mediated reactions, such as the reactions of the phosphagen system, the rate of product formation is greatly influenced by the reactant concentrations. This is denoted in equations 3.1, 3.2, and 3.3 by the two-way arrow between reactants and products. For example, as ATP is hydrolyzed to yield the energy necessary for exercise (equation 3.1), there is a transient increase in ADP concentrations (as well as P_i) in the sarcolemma. This will increase the rate of creatine kinase and adenylate kinase reaction (equations 3.2 and 3.3) to replenish the ATP supply. The process will continue until the exercise ceases or the intensity reduces such that it does not deplete CP stores and thus glycolysis or the oxidative system can become the primary supplier of ATP and rephosphorylate the free creatine (equation 3.2) (37). At this point, the sarcoplasmic concentration of ATP will remain steady or increase, which will slow down or reverse the directions of the creatine kinase and adenylate kinase reactions. As a result, equations 3.1, 3.2, and 3.3 are often referred to as **near-equilibrium reactions** that proceed in a direction dictated by the concentrations of the reactants due to the law of mass action.

Glycolytic System

Glycolysis is the breakdown of carbohydrate—either as glycogen stored in the muscle or glucose delivered via the blood—to resynthesize ATP (22, 145). The process of glycolysis involves multiple enzymatically catalyzed reactions ([figure 3.2](#)). As a result, the ATP resynthesis rate during glycolysis is not as rapid as with the single-step phosphagen system; however, the capacity to produce ATP is much higher due to a larger supply of glycogen and glucose

compared to CP. As with the phosphagen system, glycolysis occurs in the sarcoplasm.

As depicted in [figure 3.2](#), **pyruvate**, the end result of glycolysis, may proceed in one of two directions:

1. Pyruvate can be converted to **lactate** in the sarcoplasm.
2. Pyruvate can be shuttled into the mitochondria.

When pyruvate is converted to lactate, ATP resynthesis occurs at a faster rate via the rapid regeneration of NAD^+ but is limited in duration due to subsequent H^+ production resulting in decreased cytosolic pH. However, the excess H^+ does not primarily result from the rapid hydrolysis of ATP during the conversion of pyruvate to lactate; it is the breakdown of ATP to ADP and P_i , which releases H^+ ions. The conversion process of pyruvate to lactate is often called **anaerobic glycolysis** (or **fast glycolysis**). However, when pyruvate is shuttled into the mitochondria to undergo the Krebs cycle, the ATP resynthesis rate is slower because of the numerous reactions but can occur for a longer duration if the exercise intensity is low enough. This process is often referred to as **aerobic glycolysis** (or **slow glycolysis**). At higher exercise intensities, pyruvate and NADH will increase above what can be handled by pyruvate dehydrogenase and will then be converted into lactate and NAD^+ . Unfortunately, because glycolysis itself does not depend on oxygen, the terms *anaerobic* and *aerobic* (or *fast* and *slow*, respectively) *glycolysis* are probably not practical for describing the processes. Nevertheless, the fate of pyruvate is ultimately controlled by the energy demands within the cell. If energy demand is high and must be transferred quickly, as in resistance training, pyruvate is primarily converted to lactate in support of anaerobic glycolysis. If energy demand is not as high and oxygen is present in sufficient quantities in the cell, pyruvate can be further oxidized in the mitochondria.

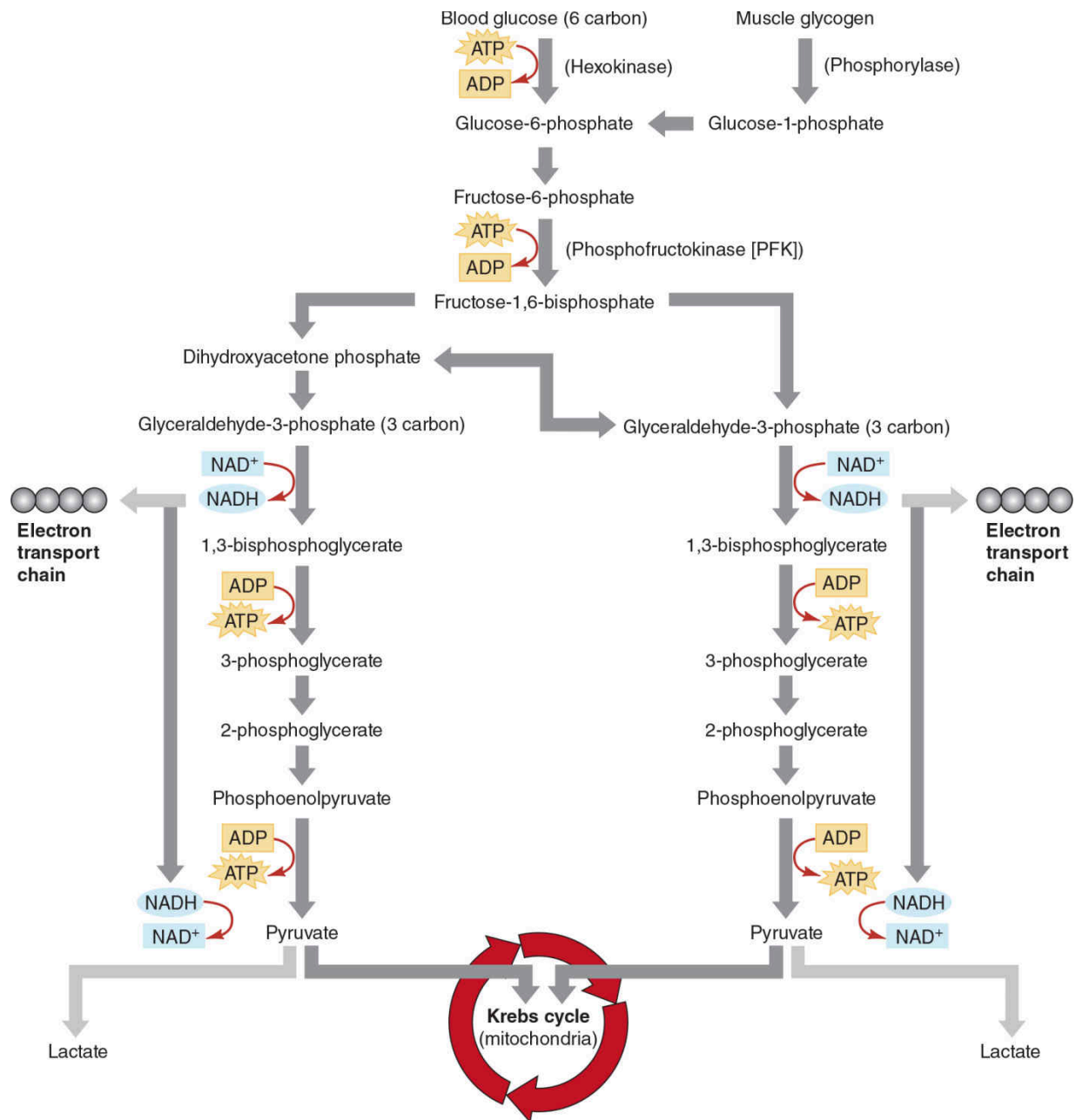


FIGURE 3.2 Glycolysis. ADP = adenosine diphosphate; ATP = adenosine triphosphate; NAD⁺, NADH = nicotinamide adenine dinucleotide.

Glycolysis and the Formation of Lactate

The formation of lactate from pyruvate is catalyzed by the enzyme lactate dehydrogenase. Sometimes, mistakenly, the end result of this reaction is said to be the formation of **lactic acid**. However, due to the physiological pH (i.e., near 7) and earlier steps in glycolysis that

consume protons (125), lactate—rather than lactic acid—is the product of the lactate dehydrogenase reaction. Although the muscular fatigue experienced during exercise often correlates with high tissue concentrations of lactate, lactate is not the cause of fatigue (22, 27, 125). Proton (H^+) accumulation during fatigue reduces the intracellular pH, inhibits glycolytic reactions, and directly interferes with muscle's excitation–contraction coupling—possibly by inhibiting calcium binding to troponin (57, 115) or by interfering with crossbridge recycling (51, 57, 79, 115, 146). Also, the decrease in pH inhibits the enzymatic turnover rate of the cell's energy systems (9, 78). Overall, this process of an exercise-induced decrease in pH is referred to as **metabolic acidosis** (125), and may be responsible for much of the peripheral fatigue that occurs during exercise (42, 156). Also, the role of metabolic acidosis in peripheral fatigue has been questioned (130), and other factors have been reported to play a prominent role in peripheral fatigue, such as an increased interstitial K^+ concentration and P_i that impairs Ca^{++} release (120, 139). However, evidence suggests that other mechanisms, such as the simple hydrolysis of ATP (125) (equation 3.1), are responsible for most of the H^+ accumulation and that lactate itself actually works to decrease metabolic acidosis rather than accelerate it (27, 125). See the sidebar *Lactic Acid Does Not Cause Metabolic Acidosis*. In fact, lactate is often used as an **energy substrate**, especially in type I and cardiac muscle fibers (10, 108, 162). It is also used in **gluconeogenesis** (i.e., the **Cori cycle**)—the formation of glucose from noncarbohydrate sources, lactate in this case—during extended exercise and recovery (19, 108).

Normally, blood and muscle have a low concentration of lactate. The reported normal range of lactate concentration in blood is 0.5 to 2.2 mmol/L at rest (67) and 0.5 to 2.2 mmol for each kilogram of **wet muscle** (muscle that has not been desiccated) (67). Lactate production increases with exercise intensity (67, 129) and appears to depend on muscle fiber type. Researchers have reported that the maximal rate of lactate production for type II muscle fibers is 0.5

mmol · g⁻¹ · s⁻¹ (46, 107) and for type I muscle is 0.25 mmol · g⁻¹ · s⁻¹ (113). The higher rate of lactate production by type II muscle fibers may reflect a higher concentration or activity of glycolytic enzymes than in type I muscle fibers (10, 122). Although the highest possible concentration of lactate accumulation is not known, severe fatigue may occur at blood concentrations between 20 and 25 mmol/L (107); one study, however, showed blood lactate concentrations greater than 30 mmol/L following multiple bouts of dynamic exercise (79). Along with exercise intensity and muscle fiber type, exercise duration (67), state of training (66), and initial glycogen levels (67) can also influence lactate accumulation.

Blood lactate concentrations reflect the net balance of lactate production versus clearance as a result of bicarbonate (HCO₃⁻) buffering. HCO₃⁻ minimizes the disrupting influence of the H⁺ on pH by accepting a proton (H⁺) and becoming carbonic acid (H₂CO₃). Carbonic acid then dissociates to H₂O and CO₂ in the lungs, where the CO₂ is exhaled. The clearance and buffering of lactate from the blood reflects a return to homeostatic resting level. Lactate can be cleared by oxidation within the muscle fiber in which it was produced, or it can be transported in the blood to other muscle fibers to be oxidized (108). Lactate can also be transported in the blood to the liver, where it is converted to glucose. This process is referred to as the **Cori cycle** and is depicted in [figure 3.3](#).

Gollnick and colleagues (67) and others (8, 72, 118) have reported that blood lactate concentrations normally return to pre-exercise values within an hour after activity, depending on the duration and intensity of exercise, training status, and type of recovery (i.e., passive versus active). Light activity during the postexercise period has been shown to increase lactate clearance rates (55, 67, 72, 80, 118). For example, an active recovery following a 200-yard (182.9 m) maximal-effort swim resulted in the greatest lactate clearance in comparison to a passive recovery in competitive swimmers (72). In addition, both aerobically trained (67) and anaerobically trained (62) athletes have faster lactate clearance

rates than untrained people. Peak blood lactate concentrations occur approximately 5 minutes after the cessation of exercise (67), a delay frequently attributed to the time required to buffer and transport lactate from the tissue to the blood (95). Hence, lactate clearance should be recognized positively as an indirect marker for recovery from metabolic acidosis as the system returns to metabolic homeostasis and not as waste product of metabolism.

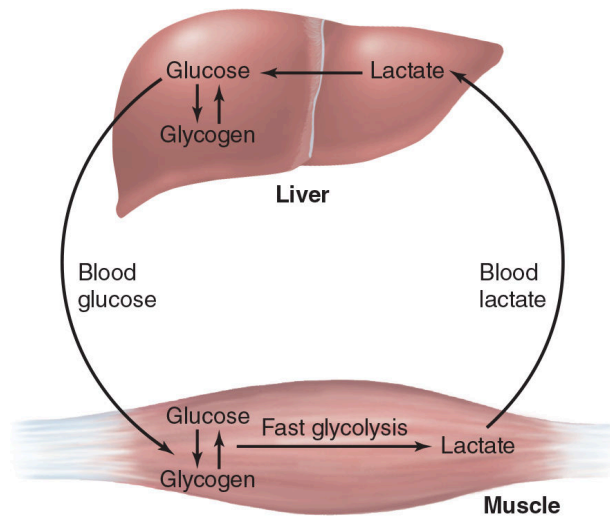


FIGURE 3.3 The Cori cycle.

Blood lactate accumulation is greater following high-intensity, intermittent exercise (e.g., resistance training and sprints) than following lower-intensity, continuous exercise (79, 103, 152). However, trained people experience lower blood lactate concentrations than untrained people when exercising at an absolute workload (same resistance) (66, 90, 143). This indicates that resistance training results in alterations in lactate response similar to those from aerobic endurance training (67, 91, 143). These alterations include a lower blood lactate concentration at a given workload in trained individuals and higher blood lactate concentrations in trained individuals during maximal exercise (67, 90, 143).

The net reaction for glycolysis when pyruvate is converted to lactate may be summarized as follows:



Lactic Acid Does Not Cause Metabolic Acidosis

Lactic acidosis is a common misnomer, because lactic acid is inaccurately believed to cause the burning sensations experienced with muscle fatigue during high-intensity exercise. This is based on the assumption that there is an immediate dissociation of lactic acid into lactate and H^+ when produced by glycolysis in skeletal muscle (1, 60, 125). However, the phosphoglycerate kinase reaction of glycolysis involves the transfer of a phosphate leaving a carboxylate (COO^-) group (105). Thus, as shown in [figure 3.4](#), no proton (H^+) exists to dissociate from lactate (60, 125).

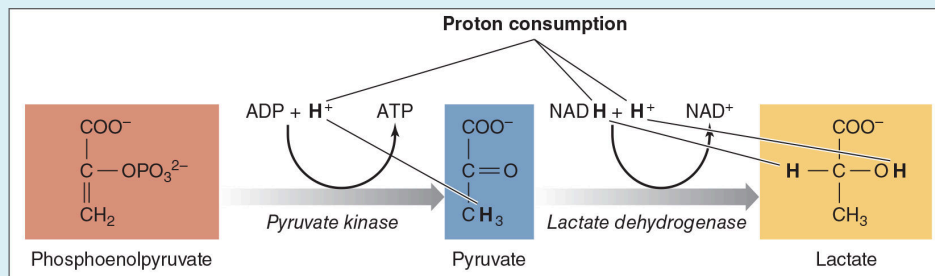


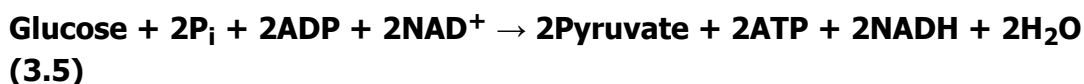
Figure 3.4 The phosphoglycerate kinase reaction of glycolysis, illustrating that there is no proton (H^+) to dissociate from lactate.

Furthermore, the lactate dehydrogenase reaction itself consumes protons, which alkalizes the cell (60, 125)—quite the opposite of acidosis. In fact, Busa and Nuccitelli (27) stated, "ATP hydrolysis, not lactate accumulation, is the dominant source of the intracellular acid load ..." (p. 430). The take-home message from Robergs and colleagues (125) is that the hydrolysis of ATP outside the mitochondria is primarily responsible for the proton (H^+) accumulation during exercise-induced metabolic acidosis—not the conversion of pyruvate to lactate as is commonly believed. Consequently, the term *lactic acidosis* is indeed a misnomer, and the term *metabolic acidosis* is recommended to describe the reduced pH within skeletal muscle during high-intensity, fatiguing exercise.

Glycolysis Leading to the Krebs Cycle

If oxygen is present in sufficient quantities in the mitochondria (specialized cellular organelles where the reactions of aerobic metabolism occur), the end product of glycolysis, pyruvate, is not converted to lactate but is transported into the mitochondria. Also transported are two molecules of reduced **nicotinamide adenine dinucleotide (NADH)** produced during glycolytic reactions (*reduced* refers to the added hydrogen). When pyruvate enters the mitochondria, it is converted to acetyl-CoA (CoA stands for coenzyme A) by the pyruvate dehydrogenase complex, resulting in the loss of a carbon as CO₂. Acetyl-CoA can then enter the Krebs cycle for further ATP resynthesis. The NADH molecules drop off H⁺ at the electron transport chain, where they can also be used to resynthesize ATP.

The net reaction for glycolysis when pyruvate is shuttled to the mitochondria may be summarized as follows:

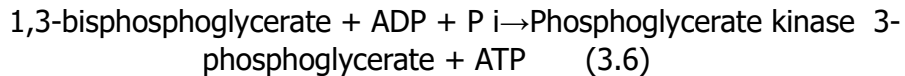


Energy Yield of Glycolysis

There are two primary mechanisms for resynthesizing ATP during metabolism:

1. Substrate-level phosphorylation
2. Oxidative phosphorylation

Phosphorylation is the process of adding a P_i to another molecule. For example, ADP + P_i → ATP is the phosphorylation of ADP to ATP. **Oxidative phosphorylation** refers to the resynthesis of ATP in the **electron transport chain (ETC)**. In contrast, **substrate-level phosphorylation** refers to the direct resynthesis of ATP from ADP during a single reaction in the metabolic pathways. To illustrate, in glycolysis two reactions result in substrate-level phosphorylation of ADP to ATP (42):



The gross number of ATP molecules that are resynthesized as a result of substrate-level phosphorylation during glycolysis is four ([figure 3.2, page 57](#)). However, the reaction that converts fructose-6-phosphate to fructose-1,6-bisphosphate (catalyzed by the enzyme **phosphofructokinase [PFK]**) in glycolysis requires the hydrolysis of one ATP molecule. In addition, there are two possible sources of glucose: blood glucose and muscle glycogen. When blood glucose enters the muscle cell, it must be phosphorylated to remain in the cell and to maintain the glucose concentration gradient (67). The phosphorylation of one molecule of blood glucose, which is catalyzed by hexokinase, also requires the hydrolysis of one ATP. In contrast, when muscle glycogen is broken down (i.e., **glycogenolysis**) to glucose with the help of the enzyme glycogen phosphorylase, the glucose is already phosphorylated, and it does not require the hydrolysis of ATP. The bottom line is that when glycolysis begins with one molecule of blood glucose, two ATP molecules are used and four ATP are resynthesized, which results in a net resynthesis of two ATP molecules. When glycolysis begins from muscle glycogen, only one ATP is used and four ATP are resynthesized, which yields a net resynthesis of three ATP molecules. The net yield rate of ATP resynthesis highlights the role and importance of maintaining muscle glycogen stores through proper nutrition.

Control of Glycolysis

In general, the rate of glycolysis is stimulated to increase during intense muscle actions by high concentrations of ADP, P_i , and ammonia and by a slight decrease in pH and AMP (22, 61, 142), all of which are signs of increased ATP hydrolysis and a need for energy. In contrast, glycolysis is inhibited by markedly lower pH, ATP, CP, citrate, and increased free fatty acids (22), which are usually present at rest. (Note that a slight decrease in pH increases

glycolysis, but if pH continues to decrease significantly it will inhibit the rate of glycolysis.) However, the regulation of glycolysis (109) is more specifically influenced by the concentrations and turnover rates of three crucial glycolytic enzymes: hexokinase, PFK, and pyruvate kinase. All three are regulatory enzymes in glycolysis, because each has important allosteric (meaning “other site”) binding sites. Allosteric regulation occurs when the end product of a reaction or series of reactions feeds back to regulate the turnover rate of key enzymes in the metabolic pathways. Consequently, this process is also called *end product regulation* (87) or *feedback regulation* (61). **Allosteric inhibition** occurs when an end product binds to the regulatory enzyme and decreases its turnover rate and slows product formation. In contrast, **allosteric activation** occurs when an “activator” binds with the enzyme and increases its turnover rate.

Hexokinase, which catalyzes the phosphorylation of glucose to glucose-6-phosphate, is allosterically inhibited by the concentration of glucose-6-phosphate in the sarcoplasm (61). Thus, the higher the concentration of glucose-6-phosphate, the more hexokinase will be inhibited. In addition, the phosphorylation of glucose commits it to the cell so that it cannot leave. Similarly, the PFK reaction (fructose-6-phosphate $\rightarrow$ fructose 1,6-bisphosphate) commits the cell to metabolizing glucose rather than storing it as glycogen. Phosphofructokinase is the most important regulator of glycolysis even though it is the slowest reaction of the steps that makes up glycolysis because it is the bottleneck, or the **rate-limiting step**. ATP is an allosteric inhibitor of PFK; therefore, as intracellular ATP concentrations rise, PFK activity decreases and reduces the conversion of fructose-6-phosphate to fructose 1,6-bisphosphate and, subsequently, decreases activity of the glycolytic pathway. However, AMP is an allosteric activator of PFK and a powerful stimulator of glycolysis. Moreover, the ammonia produced during high-intensity exercise as a result of AMP or amino acid deamination (removing the amine group from the amino acid molecule) can also stimulate PFK. Pyruvate kinase catalyzes the conversion of

phosphoenolpyruvate to pyruvate and is the final regulatory enzyme. Pyruvate kinase is allosterically inhibited by ATP and acetyl-CoA (the latter is a Krebs cycle intermediate) and activated by high concentrations of AMP and fructose-1,6-bisphosphate (61).

Lactate Threshold and Onset of Blood Lactate Accumulation

As exercise intensity increases, specific break points are evident in the lactate accumulation curve ([figure 3.5](#)), indirectly representing the change in metabolic acidosis (39, 99). The exercise intensity or relative intensity at which blood lactate begins an abrupt increase above the baseline concentration has been termed the **lactate threshold (LT) or (LT1)** (163). The LT represents a significantly increased reliance on the glycolytic pathway for energy production to meet demand. The LT corresponds well with the ventilatory threshold (breaking point in the relationship between ventilation and $\dot{V}O_2$) and is often used as a marker of the anaerobic threshold.

The LT typically begins at 50% to 60% of maximal **oxygen uptake** in untrained individuals and at 70% to 80% in aerobically trained athletes (29, 52). A second exponential increase in the rate of lactate accumulation has been noted at higher relative intensities of exercise. This second point of inflection has been termed the **onset of blood lactate accumulation (OBLA) or lactate threshold (LT2)** and occurs approximately when the concentration of blood lactate reaches 4 mmol/L (85, 138, 144). Alternatively, the LT2 threshold can be calculated, with the Dmax modified method reported to have the highest repeatability (81). The breaks in the lactate accumulation curve may correspond to the points at which intermediate and large motor units are recruited during increasing exercise intensities (94). The muscle cells associated with large motor units are typically type II fibers, which are particularly suited for anaerobic metabolism and lactate production.

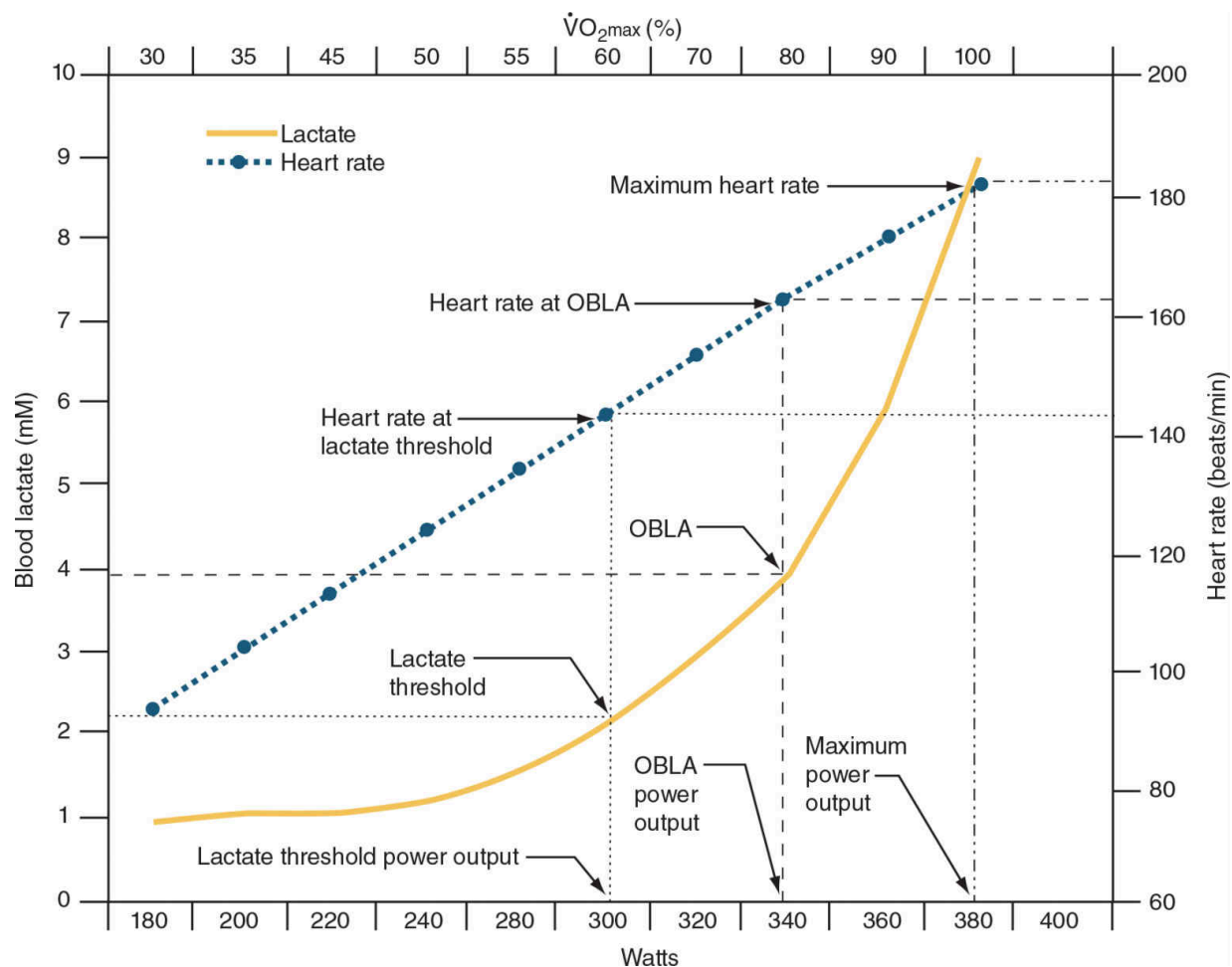


FIGURE 3.5 Lactate threshold (LT) and onset of blood lactate accumulation (OBLA).

Reprinted by permission from T.O. Bompá and G.G. Haff, *Periodization: Theory and Methodology of Training*, 5th ed. (Human Kinetics, 2009), 295.

Some studies suggest that training at intensities near or above the LT or OBLA pushes the LT and OBLA to the right (i.e., lactate accumulation occurs later at a higher exercise intensity) (39, 43). This shift probably occurs as a result of changes in hormone release, particularly reduced catecholamine release at high exercise intensities, and increased mitochondrial content that allows for greater production of ATP through aerobic mechanisms. Broadly, the training-induced adaptations result in the athlete being able to rely more on free fatty acids to meet the ATP demands of the exercise task. Thus shifting the LT or OBLA to the right allows the athlete to perform at higher percentages of maximal oxygen uptake without as

much lactate accumulation (change in metabolic acidosis) in the blood (22, 41).

The Oxidative (Aerobic) System

The oxidative system, the primary source of ATP at rest and during low-intensity activities, uses primarily carbohydrate and fats as substrates (62). Protein does not provide a significant contribution to total energy; however, the use of protein does significantly increase during long-term starvation and long bouts (>90 min) of exercise (41, 104). At rest, approximately 70% of the ATP produced is derived from fats and 30% from carbohydrate. Following the onset of activity, as the intensity of the exercise increases, there is a shift in substrate preference from fats to carbohydrate. During high-intensity aerobic exercise, almost 100% of energy is derived from carbohydrate if an adequate supply is available, with only minimal contributions from fats and protein. However, during prolonged, submaximal, steady-state work, there is a gradual shift from carbohydrate back to fats (due to glycogen depletion), and to a very small extent protein, as energy substrates (22).

Glucose and Glycogen Oxidation

The oxidative metabolism of blood glucose and muscle glycogen begins with glycolysis. If oxygen is present in sufficient quantities, the end product of glycolysis, pyruvate, is not converted to lactate but is transported to the mitochondria, where it is converted to acetyl-CoA (a two-carbon molecule), which enters the Krebs cycle, also known as the *citric acid cycle* or *tricarboxylic acid cycle* (7, 61). The Krebs cycle is a series of reactions that continues the oxidation of the substrate from glycolysis and produces two ATP indirectly from guanine triphosphate (GTP), via substrate-level phosphorylation, for each molecule of glucose ([figure 3.6](#)).

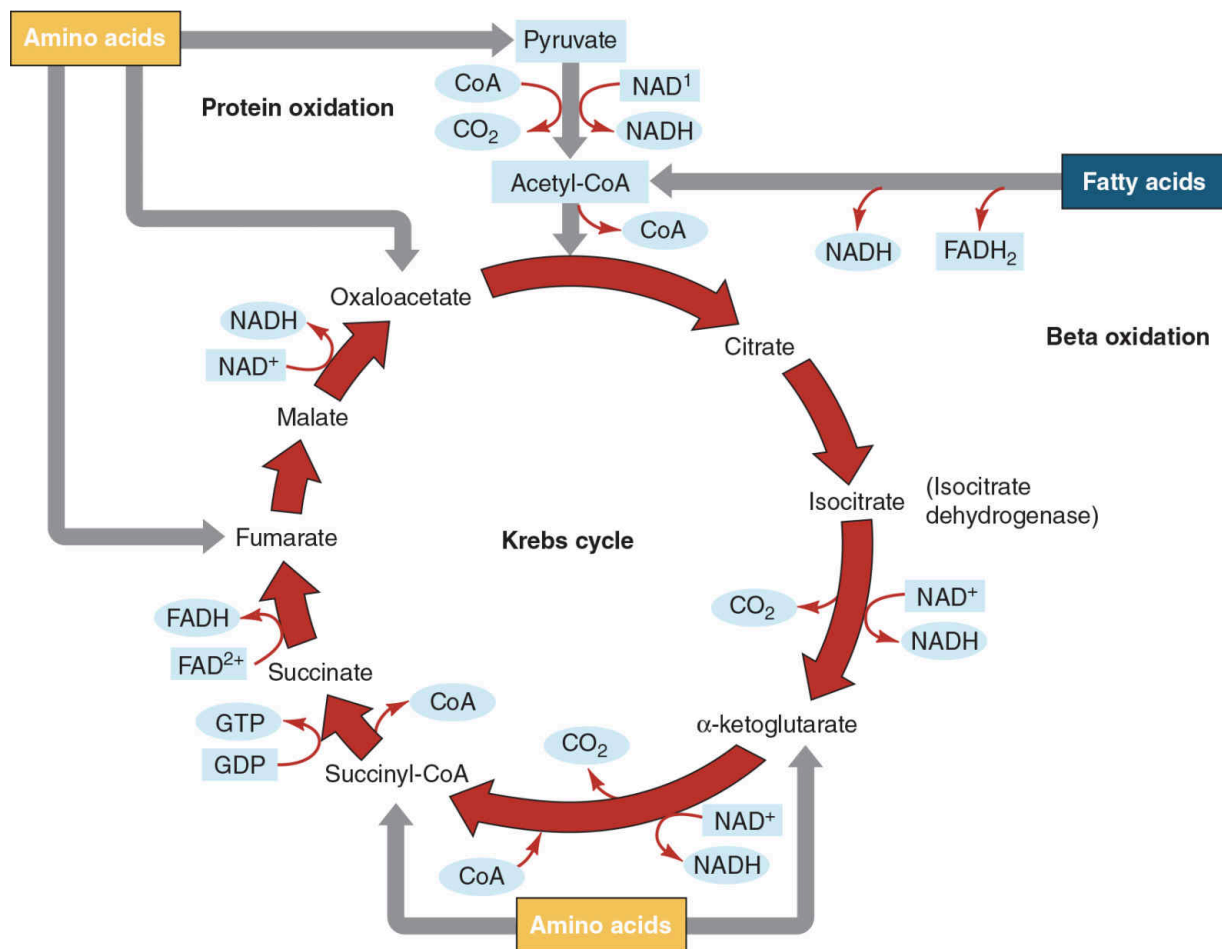


FIGURE 3.6 The Krebs cycle. CoA = coenzyme A; FAD²⁺, FADH, FADH₂ = flavin adenine dinucleotide; GDP = guanine diphosphate; GTP = guanine triphosphate; NAD⁺, NADH = nicotinamide adenine dinucleotide.

Also produced from the two pyruvate molecules subsequent to the production of one molecule of glucose are six molecules of NADH + H⁺ and two molecules of reduced **flavin adenine dinucleotide (FADH₂)**. These molecules transport hydrogen atoms to the ETC to be used to produce ATP from ADP (22, 109). The ETC uses the NADH + H⁺ and FADH₂ molecules to rephosphorylate ADP to ATP ([figure 3.7](#)).

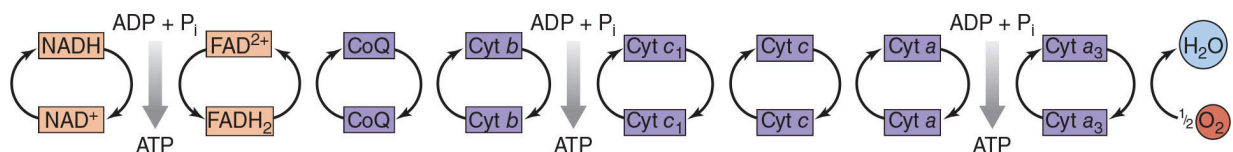


FIGURE 3.7 The electron transport chain (ETC). CoQ = coenzyme Q; Cyt = cytochrome.

The hydrogen atoms are passed down the chain (a series of electron carriers known as **cytochromes**) to form a proton concentration gradient, which provides the energy for ATP production, with oxygen serving as the final electron acceptor (resulting in the formation of water). Due to NADH + H⁺ and FADH₂ entering the ETC at different sites, they differ in the amount of ATP produced. One molecule of NADH can produce 2.5 molecules of ATP, whereas one molecule of FADH₂ can only produce 1.5 molecules of ATP (however, traditional biochemistry often cites that NADH yields 3 ATP and FADH₂ yields 2 ATP [34a]). The production of ATP during this process is referred to as *oxidative phosphorylation*. The oxidative system, beginning with glycolysis and including the Krebs cycle and ETC, results in the production of approximately 32 ATP from the degradation of one molecule of blood glucose (12a, 117a). However, if the initiation of glycolysis is muscle glycogen, the net ATP production is slightly higher at 34 ATP, using the malate-aspartate shuttle system or 32 ATP using the glycerol 3-phosphate shuttle system because the first ATP consuming step in glycolysis is bypassed (12a, 117a). Nevertheless, oxidative phosphorylation accounts for over 90% of ATP synthesis compared to substrate-level phosphorylation, which demonstrates the capacity of energy transfer by the oxidative system. See [table 3.1](#) for a summary of these processes.

TABLE 3.1 Total Energy Yield From the Oxidation of One Glucose Molecule

Process	Net ATP production
---------	--------------------

Process	Net ATP production
GLYCOLYSIS	
Substrate-level phosphorylation (4 ATP produced / 2 ATP consumed)	2*
Oxidative phosphorylation: 2 NADH (2.5 ATP each)	5
PYRUVATE OXIDATION	
2 NADH (2.5 ATP each)	5
KREBS CYCLE (2 ROTATIONS THROUGH THE KREBS CYCLE PER GLUCOSE)	
Substrate-level phosphorylation	2
Oxidative phosphorylation: 6 NADH (2.5 ATP each)	15
Via GTP: 2 FADH ₂ (1.5 ATP each)	3
TOTAL	32

*Glycolysis consumes 2 ATP (if starting with blood glucose), so net ATP production is $34 - 2 = 32$. This figure may also be reported as 30 ATP or as high as 36 depending on which shuttle system is used to transport the NADH to the mitochondria. ATP = adenosine triphosphate; FADH₂ = flavin adenine dinucleotide; GTP = guanine triphosphate; NADH = nicotinamide adenine dinucleotide.

Fat Oxidation

The oxidative energy system can alternatively use fats. Triglycerides stored in fat cells can be broken down by an enzyme, hormone-sensitive lipase, to produce free fatty acids and glycerol. The breakdown releases a portion of the total free fatty acids from the fat cells into the blood, where they can circulate and enter muscle fibers and undergo oxidation (89, 123). Additionally, limited quantities of triglycerides are stored within the muscle along with a form of hormone-sensitive lipase to produce an intramuscular source of free fatty acids (22, 47). Free fatty acids enter the mitochondria, where they undergo **beta oxidation**, a series of reactions in which the free fatty acids are broken down, resulting in the formation of

acetyl-CoA and hydrogen protons ([figure 3.6](#)). The acetyl-CoA is a common intermediary entering the Krebs cycle directly, and the hydrogen atoms are carried by NADH + H⁺ and FADH₂ to the ETC (12a, 117a). The result is hundreds of ATP molecules supplied by beta oxidation. For example, the breakdown of a single triglyceride molecule containing three 16-carbon chain free fatty acids (palmitic acid) can be metabolized by beta oxidation to yield over 300 ATP molecules (>100 ATP per palmitic acid). The overarching concept is that fat oxidation is capable of a tremendous capacity for ATP synthesis compared to carbohydrate and protein oxidation.

Protein Oxidation

Although not a significant source of energy for most activities, protein can be broken down into its constituent amino acids by various metabolic processes. Most of these amino acids can then be converted into glucose (in a process known as *gluconeogenesis*), pyruvate, or various Krebs cycle intermediates to produce ATP ([figure 3.6](#)). The contribution of amino acids to the production of ATP has been estimated to be minimal during short-term exercise but may contribute 3% to 18% of the energy requirements during prolonged activity (20, 140). The major amino acids that are oxidized in skeletal muscle are believed to be the **branched-chain amino acids** (leucine, isoleucine, and valine), although alanine, aspartate, and glutamate may also be used (69). The nitrogenous waste products of amino acid degradation are eliminated through the formation of urea and small amounts of ammonia (22). The elimination through formation of ammonia is significant because ammonia is toxic and is associated with fatigue.

Control of the Oxidative (Aerobic) System

The rate-limiting step (i.e., the slowest reaction) in the Krebs cycle ([figure 3.6](#)) is the conversion of isocitrate to α -ketoglutarate, a reaction catalyzed by the enzyme isocitrate dehydrogenase. Isocitrate dehydrogenase is stimulated by ADP and allosterically

inhibited by ATP. The reactions that produce $\text{NADH} + \text{H}^+$ or FADH_2 also influence the regulation of the Krebs cycle. If NAD^+ and FAD^{2+} are not available in sufficient quantities to accept hydrogen, the rate of the Krebs cycle is reduced. Also, when GTP accumulates, the concentration of succinyl CoA increases, which inhibits the initial reaction ($\text{oxaloacetate} + \text{acetyl-CoA} \rightarrow \text{citrate} + \text{CoA}$) of the Krebs cycle. The ETC is inhibited by ATP and stimulated by ADP (22). A simplified overview of the metabolism of fat, carbohydrate, and protein is presented in [figure 3.8](#).

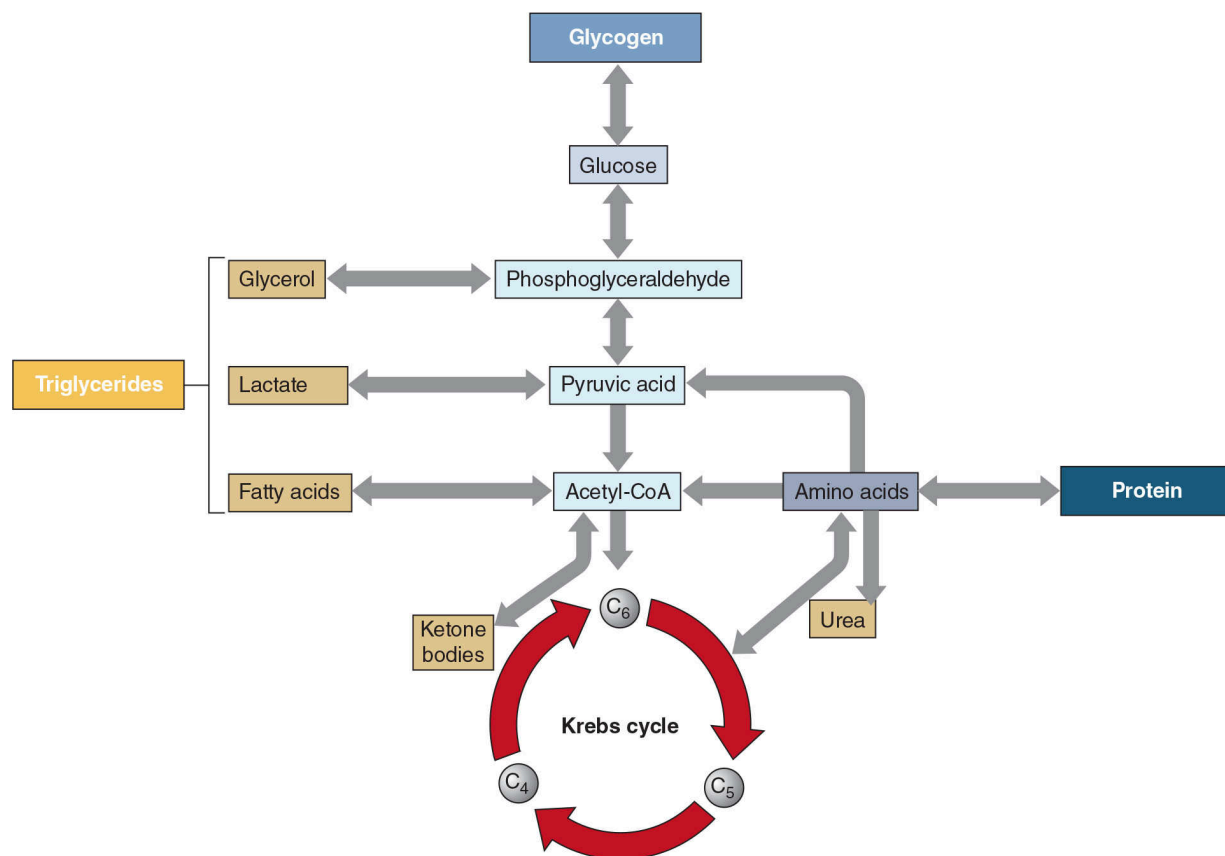


FIGURE 3.8 The metabolism of fat and that of carbohydrate and protein share some common pathways. Note that many are oxidized to acetyl-CoA and enter the Krebs cycle.

Energy Production and Capacity

The phosphagen, glycolytic, and oxidative energy systems differ in their ability to supply energy for activities of various intensities and durations ([tables 3.2](#) and [3.3](#)). Exercise intensity is often defined in

terms of mechanical power (which is the rate of doing work) as an overall measure of the external power flow required to achieve a desired level of muscular activity output (100). Activities performed at relatively high intensity for very short durations, such as resistance training, require a rapid rate of energy supply and rely almost entirely on energy provided by the phosphagen system. Activities that are of low intensity but long duration, such as marathon running, require a prolonged energy supply and rely predominantly on the energy supplied by the oxidative energy system. The primary source of energy for activities between these two extremes depends on the intensity and duration of the event ([table 3.2](#)). In general, short, high-intensity activities (e.g., high-intensity resistance training and sprinting) rely largely on the phosphagen energy system and glycolysis under anaerobic conditions whereby pyruvate is primarily converted to lactate, commonly referred to as fast glycolysis. As the intensity decreases and duration increases, the emphasis gradually shifts to slow glycolysis and the oxidative energy system (45, 131).

TABLE 3.2 Effect of Event Duration and Intensity on Primary Energy System Used

Duration of event	Intensity of event	Primary energy system
0-6 s	Extremely high	Phosphagen
6-30 s	Very high	Phosphagen and fast glycolysis
30 s to 2 min	High	Fast glycolysis
2-3 min	Moderate	Slow glycolysis and oxidative system
>3 min	Low	Oxidative system

Note: The relationships between duration, intensity, and primary energy systems used assume that the athlete strives to attain the best possible performance for a given event.

TABLE 3.3 Rankings of Rate and Capacity of ATP Production

System	Rate of ATP production	Capacity of ATP production
Phosphagen	1	5
Fast glycolysis	2	4
Slow glycolysis	3	3
Oxidation of carbohydrates	4	2
Oxidation of fats and proteins	5	1

Note: 1 = fastest/greatest; 5 = slowest/least.

In general, an inverse relationship exists between a given energy system's maximum rate of ATP production (i.e., ATP produced per unit of time) and its capacity (i.e., the total amount of ATP produced over time). The phosphagen system is capable of achieving the highest rate of ATP production, while fat oxidation through the oxidative system has the greatest capacity of ATP production. As a result, the phosphagen energy system primarily supplies ATP for high-intensity activities of short duration (e.g., 100 m sprint), the glycolytic system for moderate- to high-intensity activities of short to medium duration (e.g., 400 m dash), and the oxidative system for low-intensity activities of long duration (e.g., marathon).

The duration of the activity also influences which energy system is used. Athletic events range in duration from 1 to 3 seconds (e.g., snatch and shot put) to more than 4 hours (e.g., long-distance triathlons and ultramarathons). If an athlete makes a best effort (an effort that results in the best possible performance for a given event), the time considerations shown in [table 3.2](#) are reasonable (48, 79, 126, 146, 149).

At no time, during either exercise or rest, does any single energy system provide the complete supply of energy. During exercise, the

degree to which anaerobic and oxidative systems contribute to the energy being produced is determined primarily by the exercise intensity and secondarily by exercise duration (22, 45, 48).

The extent to which each of the three energy systems contributes to ATP production depends primarily on the intensity of muscular activity and secondarily on the duration. At no time, during either exercise or rest, does any single energy system provide the complete supply of energy.

Substrate Depletion and Repletion

Energy substrates—molecules that provide starting materials for bioenergetic reactions, including phosphagens (ATP and CP), glucose, glycogen, lactate, free fatty acids, and amino acids—can be selectively depleted during the performance of activities of various intensities and durations. Subsequently, the energy that can be produced by the bioenergetic systems is reduced. Fatigue experienced during many activities is frequently associated with the depletion of phosphagens (66, 89) and glycogen (21, 79, 92, 133). Importantly, performance is rarely limited by the depletion of substrates such as free fatty acids, lactate, and amino acids. Consequently, the **depletion** and **repletion** pattern of phosphagens and glycogen following physical activity is important in exercise and sport bioenergetics.

Phosphagens

Fatigue during exercise appears to be at least partially related to the decrease in phosphagens (i.e., ATP and CP). Phosphagen concentrations in muscle are more rapidly depleted as a result of high-intensity anaerobic exercise compared to aerobic exercise (66, 89). CP can decrease markedly (50%-70%) during the first stage of high-intensity exercise of short and moderate duration (5-30 s) and can be almost completely depleted as a result of very intense exercise to exhaustion (86, 93, 97, 110). Muscle ATP concentrations may decrease only slightly (34) or may decrease up to 50% to 60% (145) of the pre-exercise levels during experimentally induced fatigue. It should also be noted that dynamic muscle actions that produce external work use more metabolic energy and typically deplete phosphagens to a greater extent than do isometric muscle actions (18).

The intramuscular ATP concentration is largely sustained during exercise as a consequence of CP depletion and the contribution of additional ATP from the myokinase reaction and oxidation of other

energy sources, such as glycogen and free fatty acids. Postexercise phosphagen repletion can occur in a relatively short period, with complete ATP resynthesis occurring within approximately 3 to 5 minutes, and CP being fully resynthesized within 8 minutes (75, 89). Note that half of the CP stores are replenished in 30 seconds, allowing for repeated maximal contractions interspersed with brief rest intervals. While repletion of phosphagens is largely accomplished as a result of aerobic metabolism (75), glycolysis can contribute to recovery after high-intensity exercise (29, 40).

The effects of training and supplementation on resting concentrations of phosphagens is an ongoing area of investigation. Aerobic endurance training may increase resting concentrations of phosphagens (49, 99) and decrease their rate of depletion at a given absolute submaximal power output (33, 99) but not at a relative (percentage of maximum) submaximal power output (33). Although researchers have reported indications of increased resting concentrations of phosphagens (12, 127), short-term (eight weeks) studies of sprint and six months of resistance or explosive training have not shown alterations in resting concentrations of phosphagens (11, 16, 147, 150). However, total phosphagen content can be larger following sprint training due to increases in muscle mass (150). Resistance training has been shown to increase the resting concentrations of phosphagens in the triceps brachii after five weeks of training (106). The increases in phosphagen concentration are likely due to selective hypertrophy of type II fibers, which are reported to contain a higher phosphagen concentration than type I fibers (105).

Glycogen

Limited stores of glycogen are available for exercise. Approximately 300 to 400 grams (11-14 oz) of glycogen are stored in the body's total muscle and about 70 to 100 grams (2-4 oz) in the liver (137). Resting concentrations of liver and muscle glycogen can be influenced by training and dietary manipulations (56, 137). Research

suggests that both anaerobic training, including sprinting and resistance training (16, 106), and aerobic endurance training (64, 65) can increase resting muscle glycogen concentration with concomitantly appropriate nutrition.

The rate of glycogen depletion is related to exercise intensity (137). Muscle glycogen is a more important energy source than liver glycogen during moderate- and high-intensity exercise. Liver glycogen appears to be more important during low-intensity exercise, and its contribution to metabolic processes increases with duration of exercise. Increases in relative exercise intensity of 50%, 75%, and 100% of maximal oxygen uptake result in increases in the rate of muscle glycogenolysis (the breakdown of glycogen) of 0.7, 1.4, and 3.4 mmol · kg⁻¹ · min⁻¹, respectively (133). At relative intensities of exercise above 60% of maximal oxygen uptake, muscle glycogen becomes an increasingly important energy substrate; the entire glycogen content of some muscle cells can become depleted during exercise (132).

Relatively constant blood glucose concentrations are maintained at very low exercise intensities (<50% of maximal oxygen uptake) because of low muscle glucose uptake. However, as exercise duration increases beyond 90 minutes, blood glucose concentrations fall but rarely below 2.8 mmol/L (2). Long-duration exercise (>90 min) at higher intensities (>50% of maximal oxygen uptake) may result in substantially decreased blood glucose concentrations as a result of liver glycogen depletion. Hypoglycemic reactions may occur in some people with exercise-induced blood glucose values less than 2.5 mmol/L (3, 35). Reduced liver carbohydrate stores cause a decline in blood glucose to approximately 2.5 to 3.0 mmol/L, resulting in diminished carbohydrate oxidation and eventual exhaustion (32, 35, 137). However, before exhaustion occurs, it is important to note that other ATP sources are converted to glucose through gluconeogenesis, with free fatty acids becoming the predominant fuel source.

Very high-intensity, intermittent exercise, such as high-intensity resistance training, can cause substantial depletion of muscle glycogen (decreases of 20%-60%) with relatively few sets (low total volume) (100, 126, 146, 148). Although phosphagens may be the primary limiting factor during resistance exercise with high resistance and few repetitions or few sets, muscle glycogen may become the limiting factor for resistance training with larger volumes of work accumulated using many sets and repetitions (126). This type of exercise could cause selective muscle fiber glycogen depletion (more depletion in type II fibers), which can also limit performance (50, 126). As with other types of dynamic exercise, the rate of muscle glycogenolysis during resistance exercise depends on intensity (i.e., the greater the intensity, the faster the rate of glycogenolysis). However, it appears that when the total work performed is equal, the absolute amount of glycogen depletion is similar, regardless of the resistance training session's intensity (69, 126).

Repletion of muscle glycogen during recovery is related to postexercise carbohydrate ingestion. Repletion appears to be optimal if 0.7 to 3.0 grams of carbohydrate per kilogram of body weight is ingested every 2 hours following exercise (56, 137). This level of carbohydrate consumption can maximize muscle glycogen repletion at 5 to 6 mmol/g of wet muscle mass per hour during the first 4 to 6 hours following exercise. Muscle glycogen may be completely replenished within 24 hours, provided that sufficient carbohydrate is ingested (56, 137). However, if the exercise has a high eccentric component (associated with exercise-induced muscle damage), more time may be required to completely replenish muscle glycogen (121, 164).

Similar Response in Younger and Older Resistance-Trained Male Muscle Oxidative

Capacity Following Endurance Training

Hendrickse and colleagues (77) tested the hypothesis that superimposing endurance training in both older and younger highly resistance-trained men would lead to an increase in fiber oxidative capacity and a decrease in fiber size. Eight highly resistance-trained younger (mean age = 28 years) and seven older (mean age = 67 years) men participated in a 10-week cycle endurance-training program consisting of five 6-minute intervals at 75% max heart rate (HR_{max}) separated by 4-minute intervals at 90% HR_{max}, superimposed on their usual resistance training program of at least 4 sessions per week (2 upper body, 2 lower body). Before and after the 10 weeks of training, $\dot{V}O_{2\text{max}}$, body fat percentage, an MRI of the upper leg, the maximal voluntary isometric contraction torque (MVC) of the knee extensor muscles, and a vastus lateralis muscle biopsy were taken. While there were expected differences in anatomical cross-sectional area of the thigh muscles, following training no relative difference was observed between younger and older men for $\dot{V}O_{2\text{max}}$ per kilogram body mass. The imposed endurance training equally increased the succinate dehydrogenase (a Krebs cycle enzyme) activity in both younger and older participants and significantly increased the number of capillaries around type I fibers. These findings suggest that incorporating endurance training into a resistance training program can promote aerobic metabolic adaptations without compromising muscle size and strength in both younger and older resistance-trained people. Extending this research in the future will help to determine whether there is a threshold for the volume and type of endurance exercise beyond which leads to compromised muscle size and strength adaptations.

Bioenergetic Limiting Factors in Exercise Performance

Factors limiting maximal performance (22, 49, 79, 88, 104, 156) must be considered in the mechanisms of fatigue experienced during exercise and training. Understanding the possible limiting factors associated with a particular athletic event is required when one is designing training programs to achieve a physiological adaptation that seeks to delay fatigue and possibly enhance performance. [Table 3.4](#) depicts examples of various limiting factors based on depletion of energy sources and increases in muscle hydrogen ions, although other potential factors have been postulated.

Glycogen depletion can be a limiting factor both for long-duration, low-intensity exercise supported primarily by aerobic metabolism and for repeated, high-intensity exercise supported primarily by anaerobic mechanisms. Of importance to resistance training, sprinting, and other primarily anaerobic activities is the effect of metabolic acidosis on limiting contractile force (53, 79, 116, 117, 125). Several other factors have been implicated in the development of muscle fatigue and may limit exercise performance, including increased intracellular P_i , ammonia accumulation, increased ADP, and impaired calcium release from the sarcoplasmic reticulum (4, 5, 131, 156, 160). Further research is needed to delineate the causes of muscular fatigue and the limiting factors in exercise performance.

TABLE 3.4 Ranking of Bioenergetic Limiting Factors

Exercise	ATP and CP	Muscle glycogen	Liver glycogen	Fat stores	Lower pH
Marathon	1	5	4-5	2-3	1
Triathlon	1-2	5	4-5	1-2	1-2
5,000 m run	1-2	3	3	1-2	1

Note: 1 = least probable limiting factor; 5 = most probable limiting factor.

Exercise	ATP and CP	Muscle glycogen	Liver glycogen	Fat stores	Lower pH
1,500 m run	2-3	3-4	2	1-2	2-3
400 m swim	2-3	3-4	3	1	1-2
400 m run	3	3	1	1	4-5
100 m run	5	1-2	1	1	1-2
Discus	2-3	1	1	1	1
Repeated snatch exercise at 60% 1RM (10 sets)	4-5	4-5	1-2	1-2	4-5

Note: 1 = least probable limiting factor; 5 = most probable limiting factor.

Oxygen Uptake and the Aerobic and Anaerobic Contributions to Exercise

Oxygen uptake (or consumption) is a measure of a person's ability to take in oxygen via the respiratory system and deliver it to the working tissues via the cardiovascular system, and the ability of working tissues (predominantly skeletal muscle) to use oxygen. During low-intensity exercise with a constant power output, oxygen uptake increases for the first few minutes until a steady state of uptake (oxygen demand equals oxygen consumption) is reached ([figure 3.9](#)) (7, 85).

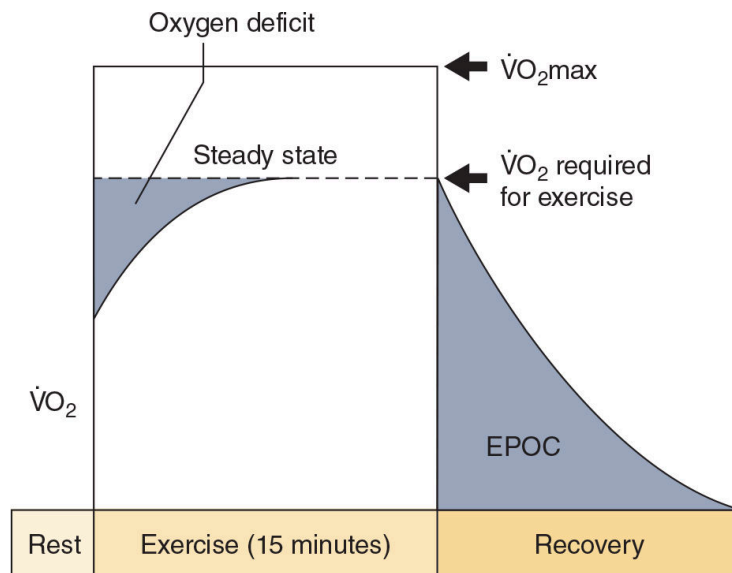


FIGURE 3.9 Low-intensity, steady-state exercise metabolism: 75% of maximal oxygen uptake ($\dot{V}O_{2max}$). EPOC = excess postexercise oxygen consumption; $\dot{V}O_2$ = oxygen uptake.

At the start of an exercise bout, however, some of the energy must be supplied through anaerobic mechanisms because the aerobic system responds slowly to the initial increase in the demand for energy (62, 155). This anaerobic contribution to the total energy cost of exercise is termed the **oxygen deficit** ([figure 3.9](#)) (85, 109). After exercise ceases, oxygen uptake remains above pre-exercise

levels for a period of time that varies according to the intensity and length of the exercise. Postexercise oxygen uptake has been termed the **oxygen debt** (85, 109), recovery O_2 (109), or the **excess postexercise oxygen consumption (EPOC)** (22). The EPOC is the oxygen uptake above resting values used to restore homeostasis to the pre-exercise condition (141). Only small to moderate relationships between the oxygen deficit and the EPOC have been observed (13, 77); the oxygen deficit may influence the size of the EPOC, but the two are not equal. The possible factors affecting the EPOC are listed in the sidebar [Excess Postexercise Oxygen Consumption Is Intensity, Duration, and Mode Dependent](#) (17, 21, 22, 58, 109).

Excess Postexercise Oxygen Consumption Is Intensity, Duration, and Mode Dependent

Excess postexercise oxygen consumption (EPOC) refers to the prolonged increase in $\dot{V}O_2$ that may be observed for hours after exercise (58).

Aerobic Exercise and EPOC (17)

- Intensity has the greatest effect on EPOC.
- The greatest EPOC values are found when both exercise intensity (i.e., >50%-60% $\dot{V}O_{2\max}$) and durations (i.e., >40 min) are high.
- Performing brief, intermittent bouts of supramaximal exercise (i.e., >100% $\dot{V}O_{2\max}$) may induce the greatest EPOC with lower total work.
- There is interindividual variability for EPOC in response to a given relative exercise stimulus.
- The effects of aerobic exercise modes on EPOC are unclear.

Resistance Exercise and EPOC (17)

- Heavy resistance exercise (i.e., three sets, eight exercises to exhaustion, 80%-90% 1RM) produces greater EPOC than circuit weight training (i.e., 4 sets, 8 exercises, 15 repetitions, 50% 1RM).
- Thus, EPOC is also intensity dependent in response to resistance training.

Factors Responsible for EPOC (17)

- Replenishment of oxygen in blood and muscle
- ATP and CP resynthesis
- Increased body temperature, circulation, and ventilation
- Increased rate of triglyceride–fatty acid cycling
- Increased protein turnover
- Changes in energy efficiency during recovery

Anaerobic mechanisms provide much of the energy for work if the exercise intensity is above the maximal oxygen uptake that a person can attain ([figure 3.10](#)). Generally, as the contribution of anaerobic mechanisms supporting the exercise increases, the exercise duration decreases (7, 68, 158, 159).

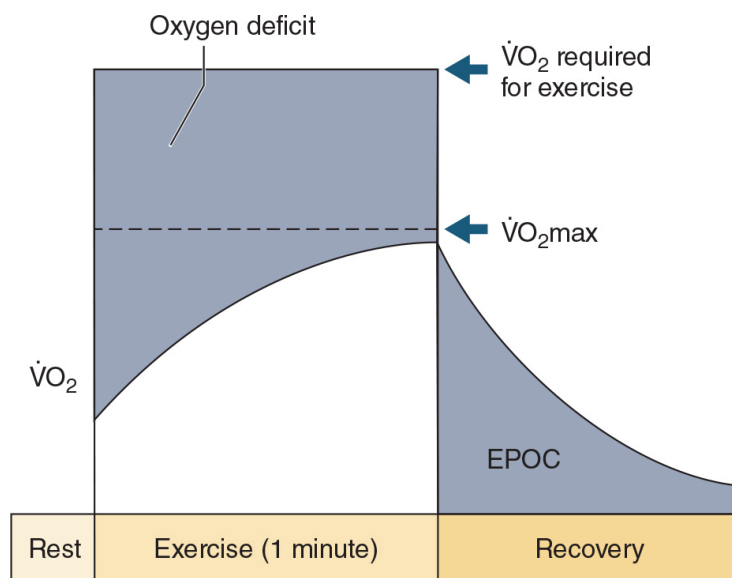


FIGURE 3.10 High-intensity, non–steady-state exercise metabolism (80% of maximum power output). The required $\dot{V}O_2$ here is the oxygen uptake that would be required to sustain the exercise if such an uptake were possible to attain. Because it is not, the oxygen deficit lasts for the duration of the exercise. EPOC = excess postexercise oxygen consumption; $\dot{V}O_{2\max}$ = maximal oxygen uptake.

The approximate contribution of anaerobic and aerobic mechanisms to maximal sustained efforts on a cycle ergometer is

shown in [table 3.5](#) (112, 151, 161). Anaerobic mechanisms are primary contributors for exercise durations up to 60 seconds, after which aerobic metabolism becomes progressively the primary energy-supplying mechanism. The contribution of anaerobic mechanisms to this type of exercise represents the maximal anaerobic capacity (111, 151).

TABLE 3.5 Contributions of Anaerobic and Aerobic Mechanisms to Maximal Sustained Efforts in Bicycle Ergometry

	0-5 s	30 s	60 s	90 s	150 s	200 s
Exercise intensity (% of maximum power output)	100	55	35	31	Not available	Not available
Contribution of anaerobic mechanisms (%)	96	75	50	35	30	22
Contribution of aerobic mechanisms (%)	4	25	50	65	70	78

Metabolic Specificity of Training

Appropriate exercise intensities and rest intervals can permit the “selection” of specific primary energy systems during training for specific athletic events (22, 109, 157). Few sports or physical activities require maximal sustained effort to exhaustion or near-exhaustion, such as competitive middle-distance sprints (400 m to 1,600 m). Most team sports and training activities produce metabolic profiles that are very similar to those of a series of high-intensity, constant- or near-constant-effort exercise bouts interspersed with rest periods, such as American football, basketball, and hockey. In this type of exercise, the required exercise intensity (work performed) that must be met during each exercise bout is much greater than that which can be sustained using aerobic energy sources alone. Increasing aerobic energy efficiency and capacity through primarily long, slow aerobic endurance training while simultaneously compromising or neglecting anaerobic energy efficiency and anaerobic capacity training is of little benefit to athletes in these sports (84, 111). For example, it would be of little benefit for a baseball player to run many long, slow miles during training rather than focusing on exercises that can concurrently improve anaerobic power and capacity using time efficient interval training.

The use of appropriate exercise intensities and rest intervals (i.e., work-to-rest ratios) allows for the “selection” of specific primary energy systems during training and, because this is more reflective of the actual metabolic demands of the sport, results in more efficient and productive regimens for specific athletic events with various metabolic demands.

Interval Training

Interval training is a method that emphasizes bioenergetic adaptations for a more efficient energy transfer within the metabolic pathways by using predetermined intervals of exercise and rest periods (i.e., **work-to-rest ratios**). Theoretically, properly spaced work-to-rest intervals allow more work to be completed at higher exercise intensities with the same or less fatigue than during continuous training at the same relative intensity. An early paper by Christensen and colleagues (31) compared the total running distance, average oxygen uptake, and blood lactate concentration during continuous running for 5 minutes and interval running totaling 30 minutes with 2:1, 1:1, and 1:2 work-to-rest ratios. Subjects were assigned a continuous running intensity (speed) that would result in fatigue within 5 minutes. At that fast pace during the continuous run, the subjects were able to complete 0.81 miles (1.3 km) before exhaustion. However, by using 2:1, 1:1, and 1:2 work-to-rest ratios and the same running intensity for a total duration of 30 minutes, the subjects were able to complete 4.14 miles (6.66 km), 3.11 miles (5 km), and 2.07 miles (3.33 km), respectively, all while working aerobic capacity in a manner similar to that in the continuous running condition. Therefore, much more distance can be accomplished at higher intensities with interval training; this concept has been established for over 45 years (31).

A series of short-term (two-week) interval training studies used six sessions of four to seven 30-second maximum cycling efforts interspaced with 4 minutes of recovery (1:8 work-to-rest ratio). These studies demonstrated improvements in muscle oxidative potential (26, 63), muscle buffering capacity (26, 63), muscle glycogen content (25, 26), and time-trial performance (25), as well as a doubled aerobic endurance capacity (26). In addition, a similar four-week interval training program exhibited increases in muscle activation and total work output (38) in trained cyclists. This highlights the concept that the modality of exercise does not seem

to influence the efficacy of using interval training for metabolic adaptations.

Few studies provide results that can be used to generate definitive guidelines for choosing specific work-to-rest ratios. One such study, however, reported aerobic and anaerobic metabolic variables, total work, and time-to-exhaustion differences between two different work-to-rest ratios in elite cyclists (119). The cyclists performed two intermittent protocols that included either a 40:20-second or a 30:30-second work-to-rest interval to exhaustion at a fixed work rate. The 40:20-second work-to-rest ratio resulted in significantly reduced total work and time to exhaustion while producing higher metabolic values ($\dot{V}O_2\text{max}$, lactate concentration, ETC). In contrast, the 30:30-second work-to-rest ratio provided sustained but slightly lower metabolic values for a considerably longer period of time. Another study manipulated the work variable via intensity and duration of the work-to-rest ratios. Wakefield and Glaister (154) reported a greater amount of time above 95% $\dot{V}O_2\text{max}$ during running at an intensity of 105% of $\dot{V}O_2\text{max}$ versus 115% of $\dot{V}O_2\text{max}$ with a work duration of 30 seconds rather than 20 and 25 seconds (rest = 20 s). When one is determining the proper work-to-rest ratio for athletes, knowledge of the time intervals, intensity of work, and recovery periods for each of the energy systems is critical to maximizing the amount of work that can be accomplished for a given exercise intensity. For example, after a bout of maximal exercise that depletes CP stores, the complete resynthesis of CP is generally 3 to 5 minutes but may take up to 8 minutes (75) even though the half time for CP resynthesis is only approximately 30 seconds, suggesting that short-duration, high-intensity exercise requires greater work-to-rest ratios due to the aerobic mechanisms that replete phosphagen stores (75).

In contrast, as the goals of training change to longer-duration, lower-intensity tasks, the durations of the work intervals can be longer; this will lengthen the rest periods and decrease the work-to-rest ratios. [Table 3.6](#) provides some general guidelines for work-to-

rest ratios that are designed to emphasize the development of specific energy systems based on the theoretical time course for metabolic system involvement and substrate recovery. However, it should be noted that more research is necessary to provide evidence-based recommendations for optimal work-to-rest ratios.

TABLE 3.6 Using Interval Training to Train Specific Energy Systems

% of maximum power	Primary system stressed	Typical exercise time	Range of work-to-rest period ratios
90-100	Phosphagen	5-10 s	1:12 to 1:20
75-90	Fast glycolysis	15-30 s	1:3 to 1:5
30-75	Fast glycolysis and oxidative	1-3 min	1:3 to 1:4
20-30	Oxidative	>3 min	1:1 to 1:3

High-Intensity Interval Training

High-intensity interval training (HIIT) involves brief repeated bouts of high-intensity exercise with intermittent recovery periods. High-intensity interval training typically incorporates either running- or cycling-based modes of exercise and is an efficient exercise regimen for eliciting cardiopulmonary (23) and metabolic and neuromuscular (24) adaptations. In fact, Buchheit and Laursen (23) stated that HIIT “is today considered one of the most effective forms of exercise for improving physical performance in athletes” (p. 314). High-intensity interval training is often discussed in terms of **duty cycles** involving a high-intensity work phase followed by a lower-intensity recovery phase. However, the duty cycle is analogous to a work bout or period as commonly used in other training regimes. It has been suggested that nine different HIIT variables can be manipulated to achieve the most precise metabolic specificity (23), including

- intensity of the active portion of each work bout,

- duration of the active portion of each work bout,
- intensity of the recovery portion of each work bout,
- duration of the recovery portion of each work bout,
- number of work bouts performed in each set,
- number of sets,
- rest time between sets,
- recovery intensity between sets, and
- mode of exercise for HIIT.

The authors (24) indicate, however, that the intensities and durations of the active and recovery portions of each work bout are the most important factors to consider. To optimize HIIT training adaptations for athletes, HIIT sessions should maximize the time spent at or near $\dot{V}O_2\text{max}$. In comparison to more general interval training, during HIIT specifically, the cumulative duration and intensity of the active portions of the work bouts should equate to several minutes above 90% of $\dot{V}O_2\text{max}$ (24).

The benefits of a HIIT protocol designed to repeatedly elicit a very high percentage of $\dot{V}O_2\text{max}$ are primarily the result of the concurrent recruitment of large motor units and near-maximal cardiac output (6). Thus, HIIT provides a stimulus for both oxidative muscle fiber adaptation and myocardial hypertrophy. Additional HIIT adaptations include increases in $\dot{V}O_2\text{max}$, proton buffering, glycogen content, anaerobic thresholds, time to exhaustion, and time-trial performance. For example, Gibala and coworkers (63) reported equivalent improvements in muscle buffering capacity and glycogen content for HIIT at 250% of $\dot{V}O_{2\text{peak}}$ during four to six 30-second cycling sprints compared to continuous cycling for 90 to 120 minutes at 65% of $\dot{V}O_{2\text{peak}}$ over six total training sessions. In addition, 750 kilojoules cycling time trials decreased in both groups by 10.1% and 7.5% in the HIIT and long, slow endurance training groups, respectively. Thus, HIIT provided performance and physiological adaptations equivalent to those of long, slow endurance training but in a time-efficient manner.

Combining Training Modalities

Some suggest that aerobic endurance training should be added to the training of anaerobic athletes to enhance recovery, because it is postulated that recovery relies primarily on aerobic mechanisms. In contrast, it is postulated that anaerobic resistance training should be added to aerobic endurance training because it can aid in preventing soft tissue injuries. However, when considering the metabolic specificity of training, thought should be given to whether the practitioner is programming combination training, cross-training, or concurrent training. These are related but distinctly different concepts when seeking to achieve metabolic specific training adaptations. Broadly, **combination training** refers to any training program that includes a mix of different types of exercises or activities. It does not necessarily imply a specific structure or timing, just that various exercise modalities are combined to achieve overall fitness improvements. For example, an 800-meter runner may include sessions of long, slow distance aerobic work alongside more event-specific distances and efforts. In contrast, **cross-training** deliberately involves incorporating different types of exercise with the same underlying metabolic adaptation profile (i.e., enhancing aerobic capacity), without overstressing the primary activity muscles (e.g., including rowing in a program for a runner), often with the aim of complementing the main sport performance. In comparison, **concurrent training** refers to the simultaneous training of both anaerobic resistance training and aerobic endurance training at multiple times within the same period, often within the same training session or day. In fact, several studies have demonstrated that recovery in anaerobic power output is related to endurance fitness (14, 15, 74). Bogdanis and colleagues (14) reported positive relationships in power recovery in the first 10 seconds of a cycling sprint, the resynthesis of PCr, and endurance fitness ($\dot{V}O_2\text{max}$). However, aerobic endurance training may reduce anaerobic performance capabilities, particularly for high-strength, high-power performance (82). This reduction in high-strength, high-power

performance capabilities could be due to an interference in anaerobic energy production capabilities due to aerobic endurance training (153). Additionally, combined anaerobic and aerobic endurance training can reduce gains in muscle girth (36, 128), maximum strength (36, 76, 128), and speed- and power-related performance (44, 73).

Even though the exact mechanism for this interference phenomenon is unknown and whether there is any threshold limit, it has been suggested that combination training and, more specifically, concurrent training may increase training volume to a level that may result in overtraining in comparison to aerobic and anaerobic training alone. Hickson and colleagues (82) provide evidence that concurrent training that includes progressive endurance running and cycling and heavy resistance training may produce a plateau effect and, ultimately, a decrease in strength gains. Specifically, it was reported that the concurrent training program of heavy resistance training combined with an endurance program provided significant improvements in squat strength during the first seven weeks of the program, followed by a two-week plateau and then a decrease in squat strength during the final two weeks of the program. The results provided evidence that continual strength gains may be inhibited with progressive concurrent endurance training such as running and cycling. Other mechanisms that have been suggested to hinder the development of strength when combined with endurance training are (a) decreasing rapid voluntary activation, (b) chronically lower muscle glycogen levels that limit intracellular signaling responses during resistance training, and (c) fiber type transition to slow-twitch fibers (114).

Alternatively, some studies and reviews indicate that the opposite holds true; these suggest that anaerobic training (resistance training) can improve low- and high-intensity exercise endurance (54, 83, 84, 136). Sedano and colleagues (134) reported improvements in performance of highly trained runners as a result of concurrent endurance, resistance, and plyometric training. There

was no reduction in $\dot{V}O_{2\max}$ over the 12 weeks in runners who participated in resistance and plyometric training. Furthermore, the concurrent training improved performance measures such as maximal strength, peak running velocity, and 3-kilometer (1.7 miles) time trial compared to endurance training alone. Thus, in highly trained runners it appears that strength training can improve performance while not hindering metabolic parameters ($\dot{V}O_{2\max}$).

Although oxidative metabolism is important for increased postexercise $\dot{V}O_2$, lactate removal, and PCr restoration from heavy anaerobic exercise (e.g., resistance training and sprint training) (135), care must be used in prescribing aerobic endurance training for anaerobic sports. In this context, it should be noted that specific anaerobic training can stimulate increases in aerobic power and enhance markers of recovery (54). Thus, it appears that extensive aerobic endurance training to enhance recovery from anaerobic events is not necessary and may be counterproductive in most strength and power sports.

Conclusion

Training programs with increased productivity can be designed through an understanding of how energy is produced during various types of exercise and how energy production can be modified by specific training regimens. Which energy system is used to supply energy for muscular contraction and to perform work is determined primarily by the intensity of exercise and secondarily by the duration of exercise. Metabolic responses and training adaptations are largely regulated by exercise characteristics (e.g., intensity, duration, and recovery intervals). How these responses and adaptations occur following physical activity forms the basis for metabolic specificity of exercise and training. This principle allows for enhanced athletic performance through the implementation of improved training programs.

KEY TERMS

- adenosine diphosphate (ADP)
- adenosine monophosphate (AMP)
- adenosine triphosphatase (ATPase)
- adenosine triphosphate (ATP)
- adenylate kinase reaction
- aerobic
- aerobic glycolysis
- allosteric activation
- allosteric inhibition
- anabolism
- anaerobic
- anaerobic glycolysis
- beta oxidation
- bioenergetics
- branched-chain amino acid
- calcium ATPase

catabolism
combination training
concurrent training
Cori cycle
creatine kinase
creatine phosphate (CP)
cross-training
cytochrome
depletion
duty cycles
electron transport chain (ETC)
endergonic reaction
energy
energy substrate
excess postexercise oxygen consumption (EPOC)
exergonic reaction
fast glycolysis
flavin adenine dinucleotide (FADH₂)
gluconeogenesis
glycogenolysis
glycolysis
glycolytic system
high-intensity interval training (HIIT)
hydrolysis
inorganic phosphate
interval training
Krebs cycle
lactate
lactate threshold (LT)
lactic acid
law of mass action
LT1
LT2
mass action effect

metabolic acidosis
metabolic specificity
metabolism
mitochondria
myokinase reaction
myosin ATPase
near-equilibrium reactions
nicotinamide adenine dinucleotide (NADH)
onset of blood lactate accumulation (OBLA)
oxidative phosphorylation
oxidative system
oxygen debt
oxygen deficit
oxygen uptake
phosphagen system
phosphocreatine (PCr)
phosphofructokinase (PFK)
phosphorylation
pyruvate
rate-limiting step
repletion
slow glycolysis
sodium-potassium ATPase
substrate-level phosphorylation
wet muscle
work-to-rest ratio

STUDY QUESTIONS

1. Which of the following best describes the role of lactate during the development of peripheral muscle fatigue?
 - a. Lactate accumulation and lactic acidosis are the primary cause of peripheral fatigue.
 - b. High tissue lactate concentrations often correlate with muscular fatigue, so they are thought to be the primary

cause of the changes in pH with high-intensity fatiguing exercise.

- c. Lactate is a by-product of when pyruvate is catalyzed, and although metabolic acidosis has a prominent role in peripheral fatigue, other factors such as an increased interstitial K^+ concentration and P_i that impairs Ca^{++} release have been reported to play a role in the development of peripheral fatigue.
 - d. The excess H^+ that causes peripheral fatigue primarily results from the rapid hydrolysis of ATP during the conversion of pyruvate to lactate.
2. What is the correct net reaction summary for glycolysis when pyruvate is converted to lactate?
- a. $Glucose + 2P_i + 2ADP \rightarrow 2Lactate + 2ATP + 2H_2O + H^+$
 - b. $Glucose + 2P_i + 2ADP \rightarrow 2Lactate + 2ATP + 2H_2O$
 - c. $Glucose + ADP + \text{creatine phosphate} \rightarrow ATP + \text{creatine}$
 - d. $Glucose + 2ADP \rightarrow Lactate + 2ATP$
3. Metabolism is the total of which four reactions?
- a. aerobic, anaerobic, oxidative, and glycolytic
 - b. catabolic, anabolic, exergonic, and endergonic
 - c. anabolic, catabolic, glycolytic, and oxidative
 - d. hydrolysis, catabolic, anabolic, and endergonic
4. Which enzyme is the most important regulator of glycolysis and why?
- a. phosphofructokinase, because it is the rate-limiting step
 - b. hexokinase, because it is the rate-limiting step
 - c. phosphofructokinase, because it is allosteric activating
 - d. pyruvate kinase, because it is the rate-limiting step
5. During exercise what are the primary and secondary determinants for which energy system is the primary contributing system?
- a. primary: intensity; secondary: type
 - b. primary: intensity; secondary: duration